

The residual bound, step by step

figures `fig_kernel_free.png`, `fig_design_range.png`; scripts listed in the Reproduction section

Symbols

τ, τ_{end}	time since onset; window end	x_{end}	$= C_2 \tau_{\text{end}}$, dimensionless window
y_i	measured rate at τ_i	N	number of samples in the window
C_1, C_2	nominal amplitude $K\dot{M}$ and exponent	$K, \dot{M}$	calibrated gain; ramp rate
W, z_{CoM}, J_P	weight, CoM height, pivot inertia	$W z_{CoM}$	$= J_P C_2^2$ (the identity)
$e_\omega(\tau)$	true rate minus the nominal	b, n	gyro bias; disturbance
C	fitted baseline (pre-onset median)	r_i	residual $y_i - \hat{\omega}_i$
$\delta t_c, \beta$	onset shift of the witness branch; its sinh coefficient, $\tanh(C_2 \delta t_c) = -\beta/C_1$	δe_ω	what the shift cannot absorb
$\rho(s)$	deviation forcing (small-angle + GE)	$\bar{\rho}$	window average bound on ρ , eq. (91)
$\dot{\rho}$	its time derivative	φ_{max}	10° design box
ω_{max}	nominal peak rate $K\dot{M} \sinh x_{\text{end}}$	l_{arm}, β_M	pivot arm; GE coefficient
ΔM_{win}	in-window moment excursion	$E(\tau)$	envelope $\bar{\rho} \sinh(C_2 \tau)/(J_P C_2)$, eq. (90)
n_{hi}	residual content above 5 Hz	κ_q, s_{med}	quiet-window shape ratio; median
$\hat{n}$	noise estimate, Sec. 6		in-window enhancement (1.60)

1 The measurement and the fit

The record in the window and the fitted model are

$$y_i = C_1 (\cosh(C_2 \tau_i) - 1) + e_\omega(\tau_i) + b + n_i, \quad (1)$$

$$\hat{\omega}_i = C_1^* (\cosh(C_2^* (\tau_i - \delta t_c^*)) - 1) + C, \quad (2)$$

and on the pre-onset segment the baseline matches the bias,

$$C \approx b. \quad (3)$$

2 The residual, and what the shift absorbs

Subtracting (2) from (1) with (3),

$$r_i = y_i - \hat{\omega}_i = \underbrace{C_1 (\cosh(C_2 \tau_i) - 1) + e_\omega(\tau_i)}_{\text{to be absorbed}} - C_1^* (\cosh(C_2^* (\tau_i - \delta t_c^*)) - 1) + n_i. \quad (4)$$

The question is how much of the braced part a member of the family can match. Why a sinh term, of all shapes? Two facts meet. First, e_ω *grows* like sinh: its envelope (90) is $E(\tau) \propto \sinh(C_2 \tau)$, because the deviation rides the instability's own growing mode — so peeling off a sinh multiple targets the dominant part of the error. Second, sinh is exactly what the family can absorb: it is the tangent direction of the onset shift, $\partial_{\delta t_c} C_1 (\cosh(C_2 (\tau - \delta t_c)) - 1)|_0 = -C_1 C_2 \sinh(C_2 \tau)$, so adding $\beta \sinh$ lands inside the family (made exact in (6)–(7)). Had the envelope been, say, τ^2 -shaped, subtracting $\beta \tau^2$ would target the error just as well and help not at all — the family could not represent it. The coincidence of the two — the error's dominant shape *is* the direction the fit can absorb — is the entire reason the measured residuals sit flat. Add and subtract that one sinh term — this *defines* δe_ω , and the equality holds at every τ , both window ends included:

$$C_1 (\cosh(C_2 \tau) - 1) + e_\omega(\tau) = \underbrace{C_1 (\cosh(C_2 \tau) - 1) + \beta \sinh(C_2 \tau)}_{\hat{h}(\tau), \text{ the witness}} + \delta e_\omega(\tau). \quad (5)$$

The witness is a member of the family in disguise. With the addition identity

$$\cosh(C_2 (\tau - \delta t_c)) = \cosh(C_2 \delta t_c) \cosh(C_2 \tau) - \sinh(C_2 \delta t_c) \sinh(C_2 \tau), \quad (6)$$

matching the cosh and sinh coefficients $[R \cosh(C_2 \delta t_c) = C_1, -R \sinh(C_2 \delta t_c) = \beta]$ gives, *exactly*,

$$\hat{h}(\tau) = R(\cosh(C_2(\tau - \delta t_c)) - 1) + (R - C_1), \quad R = \sqrt{C_1^2 - \beta^2}, \quad \tanh(C_2 \delta t_c) = -\beta/C_1, \quad (7)$$

a branch with amplitude R , onset δt_c , and a baseline offset $R - C_1$ — three numbers, all inside the family’s freedoms in (2) (the offset is absorbed by C : the witness’s baseline slot carries $b + (R - C_1)$, not b alone, and the “ $\approx$ ” of (3) is exactly this second-order constant; what it costs, and on which segment it lands, is settled in the remark closing the pre-onset section). On the campaign $\beta/C_1 \leq 0.023$, so the amplitude and baseline adjustments are second order ($|R/C_1 - 1| \leq 0.03\%$); the shift carries the first-order content. The remainder is

$$\delta e_\omega(\tau) = e_\omega(\tau) - \beta \sinh(C_2 \tau). \quad (8)$$

3 Choosing the shift: match the true curve at the window end

One free number, δt_c (equivalently β), and one choice: make the shifted branch meet the true curve at τ_{end} . Setting $\delta e_\omega(\tau_{\text{end}}) = 0$ in (8),

$$\beta = \frac{e_\omega(\tau_{\text{end}})}{\sinh(C_2 \tau_{\text{end}})}, \quad \delta t_c = -\frac{1}{C_2} \operatorname{artanh}\left(\frac{\beta}{C_1}\right) \approx -\frac{e_\omega(\tau_{\text{end}})}{C_1 C_2 \sinh(C_2 \tau_{\text{end}})}. \quad (9)$$

Since $e_\omega > 0$ (the true curve rises faster than the nominal), $\delta t_c < 0$: the branch starts *earlier* than the true onset. On the campaign $\beta/C_1 = 0.010\text{--}0.023$, i.e. $|\delta t_c| = 2\text{--}4$ ms.

Admissibility — why $|\beta| < C_1$ is guaranteed. (7) needs R real, i.e. $|\beta| < C_1$; otherwise the witness is not a cosh branch at all. This is provable, not assumed: the envelope (90) gives $|e_\omega(\tau_{\text{end}})| \leq \bar{\rho} \sinh(C_2 \tau_{\text{end}})/(J_P C_2)$, so $|\beta| \leq \bar{\rho}/(J_P C_2)$, and dividing by $C_1 = \dot{M}/W z_{CoM}$ with the identity $J_P C_2^2 = W z_{CoM}$,

$$\frac{|\beta|}{C_1} \leq \frac{\bar{\rho} C_2}{\dot{M}} \triangleq u. \quad (9a)$$

So $|\beta| < C_1$ holds whenever $\bar{\rho} < \dot{M}/C_2$ — *the ramp must sweep more moment in one e-folding time $1/C_2$ of the instability than the window-averaged remainder.* (An *e-folding* is the natural unit of an exponential process: the interval over which $e^{C_2 \tau}$ grows by a factor e , i.e. $1/C_2 \approx 0.2$ s here — the base- e analogue of a half-life. $x_{\text{end}} = C_2 \tau_{\text{end}}$ counts how many such units the window spans; this document measures time in $1/C_2$ throughout, because in that unit the same formulas give the same numbers for every vehicle and ramp rate.) A protocol that violated this would have its excitation comparable to the model error it is meant to dominate, so any meaningful design satisfies it with room: $u \leq 0.59$ at the campaign’s slowest ramp, ≤ 0.66 at the worst design-box corner of Sec. 9, and the realised $\beta/C_1 = 0.010\text{--}0.023$ sits thirty times below the ceiling. The same u is the argument of the $\operatorname{artanh}(u)/u$ factor in (21): one dimensionless number controls both the witness’s admissibility and the exactness of the shift ceiling.

What if $u \geq 1$ — ramps slower than $\bar{\rho} C_2$. Three things, in decreasing order of certainty. (i) Only the *a priori* guarantee is lost: (9a) is sufficient, not necessary, so the witness may well still be admissible — the realised β/C_1 sits some thirty times below the envelope ceiling on the campaign — but nothing proves it before the experiment any more. (ii) If $|\beta| \geq C_1$ did occur, the combination $C_1 \cosh(C_2 \tau) + \beta \sinh(C_2 \tau)$ is no longer a cosh branch: it equals $\sqrt{\beta^2 - C_1^2} \sinh(C_2 \tau + \varphi)$ (a pure exponential at $|\beta| = C_1$), and no real time shift of the family (2) draws a sinh-type curve. Step (10) then has no admissible competitor to hand to the minimiser, and the cap loses its proof — even though the Dirichlet bound (11)–(17) on $e_\omega - \beta \sinh(C_2 \tau)$ itself survives, since it never used admissibility. Moving the matching point earlier does not rescue it: the envelope of e_ω and $\sinh(C_2 \tau)$ grow in the same ratio, so the required β ceiling is the same at every matching point. (iii) Physically, $\dot{M} < \bar{\rho} C_2$ says the moment swept in one e-folding of the instability no longer dominates the model remainder — the identification concept itself, not just its certificate, is degrading; the campaign’s slowest ramp already runs at $u = 0.59$, close to that boundary. Numerically, $u = 1$ is reached at $\dot{M} = \bar{\rho} C_2 = 0.059$ N m/s for the campaign vehicle and at $\dot{M} = 0.054$ N m/s at the worst design-box corner of Sec. 9 ($m = 3.22$ kg, $l_p = 0.140$ m, $p_{\text{off}} = -0.020$ m, $z_{CoM} = 0.20$ m); a hypothetical $\dot{M} = 0.03$ N m/s ramp at that corner would have $u = 1.81$. So (9a) doubles as an analytical *lower* rate limit for the protocol, $\dot{M} > \bar{\rho} C_2 \approx 0.06$ N m/s — about half the slowest campaign rate — complementing the allocator-saturation *upper* limit on $\dot{M}$. In practice the logic runs forward, and this is the intended reading: the guarantee fails *only* there, and since $\bar{\rho}$ and C_2 need nothing but the coarse design-box ranges known before any experiment, one computes the box-worst $\bar{\rho} C_2 = 0.054$ N m/s in closed form first and then schedules the ramp rates above that floor with margin — the campaign’s $\dot{M} \geq 0.10$ N m/s is a factor ≈ 2 above it, giving $u \leq 0.66$ everywhere in the box.

Could the family be composed so the problem never arises? Partly, and where it splits is instructive. Replace the shifted-cosh family (2) by its linear span, fitting $\hat{\omega} = A(\cosh C_2\tau - 1) + B\sinh C_2\tau + C$: the witness $\omega_{\text{nom}} + \beta \sinh(C_2\tau)$ is then a member for *every* β (take $A = C_1$, $B = \beta$, $C = b$), so the argmin step (10) holds unconditionally and admissibility disappears from the residual cap altogether. The added column is no stranger: $\sinh$ is exactly the tangent direction of the time shift ($\partial_{\delta t_c}$ of the branch at $\delta t_c = 0$ is $-C_1 C_2 \sinh$, Appendix A.2), so this is the linearised version of the same estimator. The cost is one more degree of freedom in the noise term (minor) and a near-collinear pair of columns — $\cosh - 1$ and $\sinh$ agree to $O(e^{-2x_{\text{end}}})$ at the window end — so the residual stays clean while the individual coefficients A, B get noisy: the ridge of Sec. 4, resurfacing as conditioning. What *cannot* be designed away is the readout. Converting (A, B) back into an onset time needs $R = \sqrt{A^2 - B^2}$ real, i.e. $|B| < A$ — the very same condition, now appearing where it belongs. $u < 1$ is not an artifact of the cosh parametrisation; it is the statement that the data actually describes a delayed cosh onset rather than a sinh-type curve, and no reparametrisation makes data interpretable that is not. The practical split: the linear family buys an *unconditional* residual cap, and $u < 1$ then governs only whether the fitted curve has a physical onset to read off.

Then, because the minimiser in (2) is at least as good as this one particular shifted branch,

$$\boxed{\sum_{i=1}^N r_i^2 \leq \sum_{i=1}^N (\delta e_\omega(\tau_i) + n_i)^2 \implies \text{RMS}(r) \leq \text{RMS}(\delta e_\omega) + \text{RMS}(n).} \quad (10)$$

Everything now rests on one question: **how big is δe_ω ?**

4 In exactly what sense the true curve is a time-shifted nominal

The ansatz of (5) deserves a precise statement, because “the true model is the nominal, time-shifted” is true in one sense, exactly, and false in another, and the gap between the two *is* δe_ω .

The local (osculating) sense — exact at every instant. Splitting the manuscript’s Duhamel kernel (89) with the addition identity gives the deviation two running coefficients,

$$e_\omega(\tau) = A(\tau) \cosh(C_2\tau) - B(\tau) \sinh(C_2\tau), \quad A = \frac{1}{J_P} \int_0^\tau \cosh(C_2s) \rho ds, \quad B = \frac{1}{J_P} \int_0^\tau \sinh(C_2s) \rho ds, \quad (10a)$$

so the true structural curve is $(C_1 + A) \cosh(C_2\tau) - B \sinh(C_2\tau) - C_1$. Freezing the coefficients at an instant τ and applying the shift identity (6) backwards, the curve agrees there — in value *and* slope — with exactly one branch, the *osculating branch*, whose parameters are

$$R(\tau) = \sqrt{(C_1 + A(\tau))^2 - B(\tau)^2}, \quad \tanh(C_2 \delta t_{\text{loc}}(\tau)) = \frac{B(\tau)}{C_1 + A(\tau)}. \quad (10b)$$

At the onset $A = B = 0$, so $R = C_1$ and $\delta t_{\text{loc}} = 0$: **the true curve starts as the nominal, exactly.** As the window progresses the osculating parameters drift monotonically (measured on the exact tip-over: $\delta t_{\text{loc}} = 0$ at the onset, still only 2.5/0.4/0.1 ms at mid-window for $\dot{M} = 0.10/0.45/1.20$, then 546/239/85 ms at the window end, with $R(\tau_{\text{end}})/C_1 = 13.0/3.1/1.5$). The late blow-up is structural: e_ω ’s tail grows like $e^{2C_2\tau}$, faster than any branch, so the branch that hugs it must run away in amplitude and shift together — and those two parameters are ridge-degenerate, wildly different $(R, \delta t)$ pairs drawing nearly the same curve over the window. This is the same degeneracy that makes $(C_1^*, C_2^*, \delta t_c^*)$ unidentifiable from the residual.

The window (L^2) sense — the estimator’s sense. No single branch can follow the drifting profile (10b); a fit must pick one point of it. The least-squares stationarity $\langle r, \sinh \rangle = 0$ weights the choice by $\sinh^2(C_2\tau)$, which concentrates 63% of its mass in the last e-folding $1/(2C_2) \approx 90\text{--}140$ ms of the window (86% in the last two), so the fitted single shift is essentially the end-matching one of (9) — and it is *small*: $\beta/C_1 = 1.0\text{--}2.3\%$, i.e. $|\delta t_c| = 2\text{--}4$ ms on the campaign. In this sense the true curve *is* a slightly-shifted nominal: one branch, shifted by milliseconds, matches it over the whole window up to exactly δe_ω — which vanishes at both ends and sags mid-window where the drifting profile is farthest from its end value. That sag is what (11)–(17) bound.

One sign remark connects the two senses. The osculating description carries a *larger* amplitude with a *later* onset ($\delta t_{\text{loc}} > 0$); the witness of (9) carries the nominal amplitude with an *earlier* onset ($\delta t_c < 0$). Along the ridge these describe nearly the same curve — larger-and-later $\approx$ same-and-earlier — and the deployed estimator, whose amplitude is pinned to $K\dot{M}$, realises the earlier-onset representative: the sign observed on the campaign, and the sign whose M_{crit} cost (21) charges.

Remark — a perfect fit at all N samples is impossible, and that is not a defect of the fit. Two reasons, one counting and one dynamical. Counting: a branch has four free numbers $(C_1^*, C_2^*, \delta t_c^*, C)$, so it can be made to pass through at most four chosen samples; the other $N - 4$ dimensions of the sample space are out of its reach, and e_ω has components there. Dynamical, and stronger: if a branch agreed with the true curve on any subinterval, then $\delta e_\omega \equiv 0$ there, and (13) would force $\dot{\rho} \equiv 0$ on that subinterval — the true curve solves the forced equation, the branches solve the homogeneous one, and they can coincide on an interval only where the forcing vanishes. Exact agreement at $\tau_1, \dots, \tau_N$ is therefore not “difficult”; for $\dot{\rho} \neq 0$ it is unattainable in principle.

This is why the section’s claim is framed as it is. “The true curve is a time-shifted nominal” cannot be asserted exactly; asserted with the remainder, it *cannot be wrong*: eq. (5) is an identity — δe_ω is defined as whatever the shifted branch misses — and the entire scientific content lives in one place, the bound (17) on that remainder. The structure is that of Taylor’s theorem: “ f is approximately a polynomial” is a hope, “ f equals the polynomial plus a remainder obeying a stated bound” is a theorem. Here the remainder is $0.16\text{--}0.17^\circ/\text{s}$ realised, capped by (17) at $0.5\text{--}0.9^\circ/\text{s}$ from pre-experiment constants, and located — zero at both ends, sagging mid-window — so the imperfection of the time-shift description is quantified in size, ceiling and shape.

The same statement has a clean linear-algebra reading: the model family is a four-parameter manifold in the N -dimensional sample space, the true record carries a component transverse to it that no choice of parameters can reach, and δe_ω is exactly that transverse coordinate — introduced so that the description becomes an identity rather than an approximation. The Appendix makes this picture precise and connects it to the manuscript’s projector formalism.

5 How big is δe_ω : two boundary zeros and one comparison

Step 1 — δe_ω vanishes at both ends. At $\tau = 0$: $e_\omega(0) = 0$ (the deviation starts at rest) and $\sinh(0) = 0$, so $\delta e_\omega(0) = 0$. At $\tau = \tau_{\text{end}}$: zero by the choice (9).

$$\delta e_\omega(0) = \delta e_\omega(\tau_{\text{end}}) = 0. \quad (11)$$

Step 2 — the equation δe_ω obeys. The deviation dynamics of the manuscript are $J_P \ddot{e}_\varphi = W z_{CoM} e_\varphi + \rho$ with $e_\omega = \dot{e}_\varphi$. Differentiating once and using $W z_{CoM} = J_P C_2^2$,

$$\ddot{e}_\omega = C_2^2 e_\omega + \dot{\rho}/J_P. \quad (12)$$

$\sinh(C_2 \tau)$ satisfies $(\sinh)'' = C_2^2 \sinh$ — it is a solution of the same equation with zero forcing — so subtracting $\beta \sinh$ changes nothing on the right side:

$$\ddot{\delta e}_\omega = C_2^2 \delta e_\omega + \dot{\rho}/J_P, \quad \delta e_\omega(0) = \delta e_\omega(\tau_{\text{end}}) = 0. \quad (13)$$

Step 3 — comparison (this is the whole proof). Let $\bar{f}(\tau) \geq |\dot{\rho}(\tau)|$ and let w solve the same problem with forcing $-\bar{f}$:

$$\ddot{w} = C_2^2 w - \bar{f}/J_P, \quad w(0) = w(\tau_{\text{end}}) = 0. \quad (14)$$

Claim: $|\delta e_\omega(\tau)| \leq w(\tau)$. Proof in two lines: the difference $u = w - \delta e_\omega$ satisfies $\ddot{u} - C_2^2 u = -(\bar{f} + \dot{\rho})/J_P \leq 0$ with $u = 0$ at both ends — note the ≤ 0 belongs to this *combination*, at every τ , because $\bar{f} \geq |\dot{\rho}|$ makes $\bar{f} + \dot{\rho} \geq 0$. If u had a negative interior minimum, there $\ddot{u} \geq 0$ (the valley bottom bends upward: $\dot{u} = 0$ there, and $u(\tau^* + h) = u(\tau^*) + \frac{1}{2}\ddot{u}(\tau^*)h^2 + o(h^2) \geq u(\tau^*)$ for both signs of h forces $\ddot{u} \geq 0$) and $-C_2^2 u > 0$ (strictly, because $u < 0$ there — a minimum value ≥ 0 would give $-C_2^2 u \leq 0$ and no contradiction), so $\ddot{u} - C_2^2 u > 0$ — contradicting the ≤ 0 that holds everywhere. Hence no negative interior minimum exists; with $u = 0$ at both ends and u continuous, $u \geq 0$, i.e. $\delta e_\omega \leq w$; the same argument with $u = w + \delta e_\omega$ gives $\delta e_\omega \geq -w$. $\square$

Step 4 — solve (14) in closed form. For constant $\bar{f} = \sup |\dot{\rho}|$, substitution verifies

$$w(\tau) = \frac{\sup |\dot{\rho}|}{J_P C_2^2} \left(1 - \frac{\cosh(C_2(\tau - \frac{\tau_{\text{end}}}{2}))}{\cosh(x_{\text{end}}/2)} \right) \leq \frac{\sup |\dot{\rho}|}{W z_{CoM}}. \quad (15)$$

(Here $x_{\text{end}} = C_2 \tau_{\text{end}}$: at $\tau = 0$ and $\tau = \tau_{\text{end}}$ the numerator equals $\cosh(x_{\text{end}}/2)$, so the bracket vanishes at both ends; it peaks at mid-window, where the bound is $(1 - \text{sech}(x_{\text{end}}/2)) \sup |\dot{\rho}|/W z_{CoM}$.) **No exponential appears.** The envelope (90) grows like $\sinh(C_2 \tau)$ because nothing stops the unstable mode of (12); here the two boundary zeros of (11) — one free, one bought with δt_c — pin that mode down. This is what “the fit absorbs the growing part” means as an equation.

Step 5 — the forcing rate, from the same channels as $\bar{\rho}$. Differentiating the gravity remainder (84) in time gives $\dot{\rho}_{\text{grav}} = [Wl_{\text{arm}} \sin \varphi + Wz_{CoM}(\cos \varphi - 1)]\omega$, and the two bracket terms have *opposite signs*: the height term is negative and cancels against the arm term instead of adding to it. The bracket stays inside $[0, Wl_{\text{arm}} \sin \varphi]$ as long as $l_{\text{arm}} \geq z_{CoM} \tan(\varphi/2)$ — held with a factor 3.4 over the whole design box (0.090 m against at most $0.30 \tan 5^\circ = 0.026$ m) — so Wz_{CoM} drops from the bound *by sign, not by approximation*. This is the same cancellation $\bar{\rho}$ (91) already uses ($Wz_{CoM}(\sin \varphi - \varphi) \leq 0$ there), and it is exactly what the manuscript’s (85) says at $\varphi^* = 0$: $G'''(0) = W(l_p + p_{\text{off}})$, the $Wz_{CoM} \sin \varphi^*$ term vanishing. Bounding on the 10° box, with $\omega_{\text{max}} = K\dot{M} \sinh x_{\text{end}}$:

$$\sup |\dot{\rho}| \leq \underbrace{Wl_{\text{arm}} \varphi_{\text{max}} \omega_{\text{max}}}_{\dot{\rho}_2, \text{ decays like } e^{2C_2(s-\tau_{\text{end}})}} + \underbrace{|\beta_M|(\dot{M}\varphi_{\text{max}} + \Delta M_{\text{win}} \omega_{\text{max}})}_{\dot{\rho}_1, \text{ decays like } e^{C_2(s-\tau_{\text{end}})}}. \quad (16)$$

Every constant is available before the experiment: W, z_{CoM} from the scale, J_P from CAD, l_{arm} from geometry, φ_{max} the design box, β_M from the GE model, $\dot{M}$ and the window from the protocol.

What “decays like” means — the bridge from (16) to (16a). Nothing in the window decreases with time: φ and ω both grow. The decay is the same fact read *backwards from the window end* — the claim under each brace of (16) is

$$|\dot{\rho}_{\text{channel}}(s)| \leq |\dot{\rho}_{\text{channel}}(\tau_{\text{end}})| e^{kC_2(s-\tau_{\text{end}})}, \quad s \leq \tau_{\text{end}}, \quad k \in \{2, 1\},$$

where the exponential factor is < 1 because $s - \tau_{\text{end}} < 0$: at earlier times the forcing sits below its end value by at least that factor. This is the form the comparison problem of Step 6 consumes — a peak at the end with a known fall-off behind it. Divide both sides by e^{kC_2s} and the claim becomes $|\dot{\rho}(s)|/e^{kC_2s} \leq |\dot{\rho}(\tau_{\text{end}})|/e^{kC_2\tau_{\text{end}}}$ — i.e. *the ratio $|\dot{\rho}(s)|/e^{kC_2s}$ is nondecreasing in s* . That is why (16a) is a statement about ratios. And because each channel is a *product* of trajectory factors, $\dot{\rho}/e^{kC_2s}$ factors as a product of single-factor ratios ($\varphi\omega/e^{2C_2s} = [\varphi e^{-C_2s}][\omega e^{-C_2s}]$), a product of nonnegative nondecreasing functions is nondecreasing, and the exponent k in (16) is literally a *count of trajectory factors*: two in $\varphi\omega$ ($k = 2$), one in each ground-effect term ($k = 1$).

Why anchor at the end rather than the onset. Four reasons, the last the deepest. A forward envelope $|\dot{\rho}(s)| \leq |\dot{\rho}(0)|e^{kC_2s}$ has nothing to hang on: the forcing *vanishes* at onset ($\varphi = \omega = 0$), so its anchor amplitude is zero; the one usable anchor is the peak — the end value, which any bound must pay anyway and which the channel sups bound a priori. Second, the tightness lives in “how little forcing there is early”: the Dirichlet response has a boundary zero directly under the forcing peak, and exploiting that cancellation requires the envelope to encode the fall-off *behind* the peak — exactly what the backward rate is, and exactly what a constant (sup-only) envelope discards at a cost of 8–12 $\times$. Third, only the end-anchored coordinate $u = C_2(s - \tau_{\text{end}}) \in [-x_{\text{end}}, 0]$ makes the comparison problem forget the window length: as $x_{\text{end}} \rightarrow \infty$ the end neighbourhood is fixed and only the onset recedes, which is why the x_{end} -free constants $1/12$ and $1/(2e)$ exist at all; in an onset-anchored coordinate the peak would move with x_{end} and no such constants would emerge. Last, the instability has one clock, $1/C_2$, and everything that matters happens in the last few e-foldings before τ_{end} : *measured from the end, all windows look alike*. That statement is the coordinate version of the rate-flatness of the residual — the ramp rate only moves the onset distance x_{end} , and the end neighbourhood forgets x_{end} .

Step 5' — the decays marked in (16) are theorems, not observations. Along the nominal trajectory, write $a = C_2s$. By the bridge above, it suffices that each factor’s e^{C_2s} -ratio is nondecreasing; the first ratio of (16a) certifies the factor ωe^{-C_2s} , the second certifies φe^{-C_2s} , and both facts are one line each:

$$\frac{\omega_{\text{nom}}(s)}{C_1 e^{C_2s}} = \frac{(\cosh a - 1)}{e^a} = \frac{(1 - e^{-a})^2}{2} \nearrow, \quad \frac{d}{da} \left[\frac{\sinh a - a}{e^a} \right] = e^{-a}(a - 1 + e^{-a}) \geq 0, \quad (16a)$$

the first read off directly, the second because $e^{-t} \geq 1 - t$; the bracket in the second is $\varphi_{\text{nom}} C_2 e^{-C_2s}/C_1$, so $\varphi_{\text{nom}} e^{-C_2s}$ is nondecreasing too. Now every channel of $\dot{\rho}$ is a product of positive factors whose e^{C_2s} -ratios are nondecreasing, and such a product obeys its decay exactly. Arm channel:

$$\sin \varphi \omega \leq \varphi \omega = [\varphi e^{-C_2s}][\omega e^{-C_2s}] e^{2C_2s} \leq \varphi(\tau_{\text{end}}) \omega(\tau_{\text{end}}) e^{2C_2(s-\tau_{\text{end}})}.$$

(The height term, had it been kept, is $(1 - \cos \varphi)\omega \leq \frac{1}{2}\varphi^2\omega$ — *three* such factors, decaying as $e^{3C_2(s-\tau_{\text{end}})}$; it is absent from (16) by the sign cancellation of Step 5, so nothing needs charging.) Ground effect: $\dot{M}\varphi$ carries one factor (e^{C_2} class), and $\Delta M(s)\omega = \dot{M}s\omega$ carries $s/\tau_{\text{end}} \leq 1$ (also nondecreasing) times one, so it too is e^{C_2} class. For the *nominal* trajectory the lemma is therefore exact, with no safety factor.

Nominal to true. The true trajectory multiplies each factor by $1 + O(e/\omega_{\text{nom}})$. Measured on the exact nonlinear tip-over across the seven rates, the worst excess of a true ratio over its nominal bound is 2.6%, at

the slowest ramp near the window end; the 1.05 carried on $\dot{\rho}_2$ in (16) is that transfer allowance, held with double margin. The safety factor thus covers only the nominal-to-true step — the shape itself is proved.

Step 6 (optional tightening, factor 8) — use the decay of $\dot{\rho}$. The forcing is not constant: it peaks at the window end and decays backwards as marked in (16) and proved in Step 5'. So solve (14) once for each shape at unit peak — the same two-line comparison applies, and the solutions are still elementary. In the scaled variable $u = C_2(\tau - \tau_{\text{end}}) \in [-x_{\text{end}}, 0]$ — a *translation*, not a reflection: $du/d\tau = +C_2 > 0$, so u runs forward with τ and only the origin has moved to the window end (a reflection would be $C_2(\tau_{\text{end}} - \tau)$; the “backward” of Step 5' is how the envelope is *described*, not the orientation of the coordinate) — for the forcing $e^{2C_2(\tau - \tau_{\text{end}})} = e^{2u}$:

$$w_2(\tau) = \frac{\dot{\rho}_2}{J_P C_2^2} \cdot \frac{1}{3} \left[\frac{\sinh(u + x_{\text{end}})}{\sinh x_{\text{end}}} + e^{-2x_{\text{end}}} \frac{\sinh(-u)}{\sinh x_{\text{end}}} - e^{2u} \right], \quad M_2(x_{\text{end}}) \triangleq \max_{u \in [-x_{\text{end}}, 0]} \frac{1}{3} [\dots], \quad (17a)$$

and for the forcing $e^{C_2(\tau - \tau_{\text{end}})} = e^u$, which is *resonant* — it decays at the operator's own rate, so the particular solution carries a factor of u :

$$w_1(\tau) = \frac{\dot{\rho}_1}{J_P C_2^2} \left[\frac{x_{\text{end}}}{2} \frac{\sinh(u + x_{\text{end}})}{\sinh x_{\text{end}}} - \frac{(u + x_{\text{end}}) e^u}{2} \right], \quad M_1(x_{\text{end}}) \triangleq \max_{u \in [-x_{\text{end}}, 0]} [\dots]. \quad (17b)$$

No time reversal hides in (17a): substituting $u = C_2\tau - x_{\text{end}}$ back gives, in original time,

$$w_2(\tau) = \frac{\dot{\rho}_2}{J_P C_2^2} \cdot \frac{1}{3} \left[\frac{\sinh(C_2\tau)}{\sinh x_{\text{end}}} + e^{-2x_{\text{end}}} \frac{\sinh(C_2(\tau_{\text{end}} - \tau))}{\sinh x_{\text{end}}} - e^{2C_2(\tau - \tau_{\text{end}})} \right],$$

which vanishes at $\tau = 0$ ($0 + e^{-2x_{\text{end}}} - e^{-2x_{\text{end}}}$) and at $\tau = \tau_{\text{end}}$ ($1 + 0 - 1$), with the forcing term growing forward in τ to its peak at the end — exactly the orientation of the physical window. (Backward time $v = \tau_{\text{end}} - \tau$ would give the mirror parametrisation $w(\tau_{\text{end}} - v)$ with forcing e^{-2C_2v} — the same function, described from the other end.)

Where the two forms come from — construction, not guess. In the scaled variable the comparison problem reads $w'' - w = -\kappa g(u)$, $w(-x_{\text{end}}) = w(0) = 0$, with $\kappa = \dot{\rho}_i/(J_P C_2^2)$, one shape g at a time. (One label trap, worth naming: these are the *same* two conditions as $w(\tau=0) = w(\tau_{\text{end}}) = 0$, relabelled — since the origin of u sits at the window end, the token “0” names the *onset* in τ -labels but the *window end* in u -labels: $u = -x_{\text{end}}$ is $\tau = 0$, and $u = 0$ is $\tau = \tau_{\text{end}}$. Two physical pins, onset and end, in either language.) Three moves each. *Particular solution:* for $g = e^{2u}$ try Ae^{2u} : $(2^2 - 1)A = -\kappa$, so $A = -\kappa/3$ — the $1/3$ is the distance to resonance. For $g = e^u$ the same try fails, $(1^2 - 1)A = 0$: the forcing decays at the operator's own rate, so carry a factor u , $w_p = -(\kappa/2)ue^u$, since $(ue^u)'' - ue^u = 2e^u$ — this extra u is why $M_1 > M_2$. *Homogeneous corrections in the endpoint basis:* $\sinh(u + x_{\text{end}})$ vanishes at $u = -x_{\text{end}}$ and $\sinh(-u)$ vanishes at $u = 0$, so each boundary condition sets one coefficient without disturbing the other: $u = 0$ gives $a \sinh x_{\text{end}} = \kappa/3$, $u = -x_{\text{end}}$ gives $b \sinh x_{\text{end}} = (\kappa/3)e^{-2x_{\text{end}}}$ — that is (17a); for e^u , collecting the two u -terms into $-(u + x_{\text{end}})e^u/2$ makes the zero at $u = -x_{\text{end}}$ visible — that is (17b). *Verification by substitution:* the $\sinh$'s are annihilated by the operator, the exponentials return the forcing, and both brackets vanish at both ends — the two boundary zeros of (14). Their maxima barely depend on x_{end} and tend to simple constants,

$$M_2(x_{\text{end}}) \rightarrow \frac{1}{12}, \quad M_1(x_{\text{end}}) \rightarrow \frac{1}{2e}, \quad \text{so} \quad |\delta e_\omega| \leq \frac{M_2 \dot{\rho}_2 + M_1 \dot{\rho}_1}{W z_{CoM}}. \quad (17)$$

The limits are one-line maximisations: as $x_{\text{end}} \rightarrow \infty$, $\sinh(u + x_{\text{end}})/\sinh x_{\text{end}} \rightarrow e^u$ and the $e^{-2x_{\text{end}}}$ term dies, so the bracket of (17a) tends to $(e^u - e^{2u})/3 = (v - v^2)/3$ with $v = e^u \in (0, 1]$, maximised at $v = \frac{1}{2}$ to $1/12$ ($\ln 2$ e-foldings before the window end); the bracket of (17b) tends to $-(u/2)e^u$, maximised at $u = -1$ to $1/(2e)$ (one e-folding before the end). (Over the campaign range $x_{\text{end}} = 1.9\text{--}3.7$: $M_2 = 0.074\text{--}0.083$, $M_1 = 0.141\text{--}0.181$; the script evaluates (17a)–(17b) exactly.) The interpretation is the same as the manuscript's reduction factors $1/7$ and $1/5$: the forcing is largest exactly where the pinned response is smallest — at the window end the boundary zero sits right under the peak of the load.

A one-line version, in the spirit of (85). The limits are also *ceilings*: $M_2(x_{\text{end}})$ and $M_1(x_{\text{end}})$ approach $1/12$ and $1/(2e)$ from below (verified numerically over $x_{\text{end}} \in [0.05, 12]$ to grid precision), so (17) may be quoted x_{end} -free,

$$|\delta e_\omega| \leq \frac{\dot{\rho}_2/12 + \dot{\rho}_1/(2e)}{W z_{CoM}},$$

with no M curves, no window length, and no C_2 beyond what sits inside $\dot{\rho}_{1,2}$ — exactly as (85) trades the exact remainder for $\frac{1}{2}G''(0)\varphi^2$. Rerun on the 140 runs, the constant version costs at most 15% on the worst

run (0.2–7% in the per-rate mean, largest at the fastest ramps where x_{end} is smallest), keeps 140/140 coverage, and the worst-run used fraction rises only to 0.86. Note where the simplification is *not* available: the channel amplitudes $\dot{\rho}_{1,2}$ are already the $\varphi^* = 0$ worst-case Taylor coefficients of (85) — nothing further to drop there — and the time *shape* behind M_2, M_1 (the forcing concentrated at the window end as $e^{2C_2(s-\tau_{\text{end}})}$) cannot be dropped at all: against a constant-in-time forcing the comparison factor is $1 - \text{sech}(x_{\text{end}}/2) = 0.4\text{--}0.9$, a factor 5–10 above $1/12$, which is the loose bound this document exists to escape. The Taylor step simplifies the *amplitude*; the tightness lives in the *shape*.

6 The pre-onset segment

Where the segment comes from. By (9), $\delta t_c < 0$: the witness branch starts $|\delta t_c|$ before the true onset. On the segment $\tau \in [\delta t_c, 0]$ — length $|\delta t_c|$ — the record is still at its baseline, but the branch has already been running for $u = \tau - \delta t_c \in [0, |\delta t_c|]$ since its own onset, so the mismatch there is the branch itself:

$$\text{mismatch}(u) = C_1 (\cosh(C_2 u) - 1), \quad u \in [0, |\delta t_c|]. \quad (18a)$$

The residual sum runs over this segment too, so the witness’s cost in (10) gains its square. Normalising by the main window length τ_{end} (conservative: the extended window is longer, so dividing by the shorter τ_{end} can only enlarge the term),

$$\Delta_{\text{pre}}^2 = \frac{1}{\tau_{\text{end}}} \int_0^{|\delta t_c|} [C_1 (\cosh(C_2 u) - 1)]^2 du. \quad (18b)$$

The segment’s *noise* adds no separate term: $\hat{n}$ is an RMS density, unchanged by a slightly longer window.

The integral is elementary. Expand the square with $\cosh^2 a = \frac{1}{2}(1 + \cosh 2a)$:

$$(\cosh a - 1)^2 = \cosh^2 a - 2 \cosh a + 1 = \frac{3}{2} + \frac{1}{2} \cosh(2a) - 2 \cosh a, \quad (18c)$$

and integrate term by term with $a = C_2 u$ ($\frac{3}{2} \rightarrow \frac{3u}{2}$; $\frac{1}{2} \cosh(2C_2 u) \rightarrow \sinh(2C_2 u)/(4C_2)$; $-2 \cosh(C_2 u) \rightarrow -2 \sinh(C_2 u)/C_2$):

$$\Delta_{\text{pre}}^2 = \frac{C_1^2}{\tau_{\text{end}}} \left[\frac{3u}{2} + \frac{\sinh(2C_2 u)}{4C_2} - \frac{2 \sinh(C_2 u)}{C_2} \right]_0^{|\delta t_c|}. \quad (18)$$

Check, and why the term is small. Expanding the bracket in powers of u , the u and u^3 orders cancel identically and the first surviving order is

$$\int_0^{|\delta t_c|} (\cosh(C_2 u) - 1)^2 du \approx \frac{C_2^4 |\delta t_c|^5}{20},$$

fifth order in the shift, because the branch leaves the baseline only quadratically ($\cosh - 1 \approx C_2^2 u^2/2$, squared $\rightarrow u^4$). Bounding instead by (segment sup) $\times$ (length) would give $|\delta t_c|^5/4$ — the exact integral is a factor $\sqrt{5} \approx 2.2$ tighter. (The branch amplitude is $R \leq C_1$; using C_1 in (18) errs on the high side.) Evaluated with the a priori $|\delta t_c|$ from (9) and $e_\omega(\tau_{\text{end}}) \leq E(\tau_{\text{end}})$: $\Delta_{\text{pre}} = 0.036^\circ/\text{s}$ at $M = 0.10$, 0.011 at 0.20 , and below 0.006 beyond.

The baseline constant $R - C_1$ — both bookkeepings. One witness must serve every sample the minimiser’s cost touches, and its baseline slot holds a single number, so the constant $R - C_1$ of (7) must land somewhere. There are two consistent choices. (a) Take (3) literally, $C = b$: the flat pre-onset samples then cost nothing beyond noise, (18a) holds as written (amplitude $R \leq C_1$), and the post-onset residual gains the constant, $r = \delta e_\omega + n + (C_1 - R)$. (b) Take $C = b + (R - C_1)$: the post-onset identity is exactly (5), $r = \delta e_\omega + n$, and the constant moves to the flat stretch, where the length fraction of that stretch in the window dilutes it further. Either way the price is the same number,

$$C_1 - R = C_1 \left(1 - \sqrt{1 - (\beta/C_1)^2} \right) \leq \frac{1}{2} C_1 (\beta/C_1)^2,$$

second order in β/C_1 . Realised ($\beta/C_1 \leq 0.023$) it is below $0.004^\circ/\text{s}$ on every run — two orders under $\hat{n}$ and below even Δ_{pre} — which is why (10) and (20) carry it silently; the pre-onset median cannot even distinguish b from $b + (R - C_1)$ at that size. At the a priori envelope ceiling it is not invisible: $u = 0.59$ at the slowest ramp gives $C_1(1 - \sqrt{1 - u^2}) \approx 0.2^\circ/\text{s}$, about a fifth of the model term there. Adding it to the cap only loosens the bound, so the 140/140 check is unaffected either way, and the measured β/C_1 , thirty times below the envelope, confirms the silent treatment a posteriori.

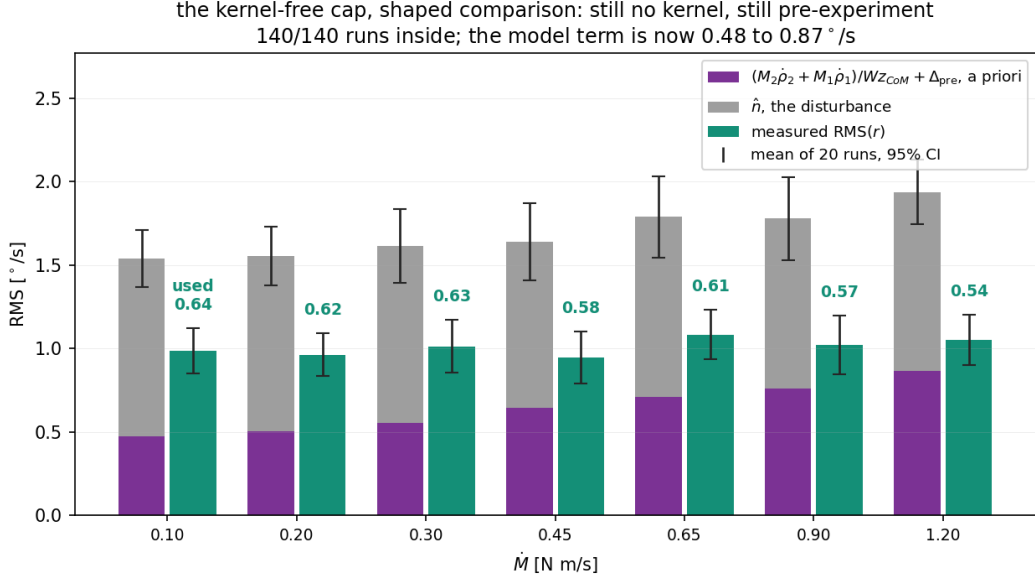


Figure 1: Equation (20) against the campaign: means over the twenty runs at each rate, 95% t intervals. Purple is the model term of (17)+(18), grey the noise estimate (19), green the measured residual. All 140 runs are inside; the used fraction is 0.54–0.64 and the cap is flat, like the residual.

7 The noise term

Above 5 Hz the model can produce nothing, so n_{hi} is measured from the record; the full band is estimated through the quiet-window shape ratio,

$$\hat{n} = \text{RMS}(n_{hi}) \sqrt{1 + (s_{med} \kappa_q)^2}, \quad (19)$$

with $s_{med} = 1.60$ the campaign-median in-window enhancement and κ_q the run's own quiet ratio (campaign median 0.37 where no quiet stretch exists).

The cutoff is not delicate, and $k\sigma$ is what (19) already is. Structurally (19) reads $k \cdot \sigma$ with $\sigma = \text{RMS}(n_{hi})$ — measured *in the window*, in a band the model cannot contaminate — and $k = \sqrt{1 + (s_{med} \kappa_q)^2} \approx 1.6$ –1.9. The split frequency only defines that band, and any choice works that sits well above the model's spectral content $C_2/2\pi = 0.6$ –0.9 Hz and leaves spectral bins on both sides ($f_c \gtrsim 2/(NT_s) \approx 3$ Hz for the shortest windows): recomputing the entire check at $f_c = 2, 3$ and 5 Hz gives identical results (140/140 inside, mean used 0.57–0.58), and only past 8 Hz does the shrinking high band make the estimate noisy (133/140 at 8, 129/140 at 10). The invariance is expected: $\hat{n}$ estimates the total $\text{RMS}(n)$, which does not depend on where the record is split. What does *not* work is the simpler $k\sigma$ with σ from the pre-onset segment: the in-window disturbance is contact-borne — pivoting on one gear edge, not gyro noise — and runs 2 to 50 times the pre-onset floor with no usable floor at all on 25 of the 140 runs, so the one campaign constant that covers the spread, $k = 49$, collapses the used fraction from 0.58 to 0.16.

8 The check on all 140 runs

$$\text{RMS}(r) \leq \underbrace{\frac{M_2 \dot{\rho}_2 + M_1 \dot{\rho}_1}{W z_{CoM}} + \Delta_{pre}}_{\text{model, before the experiment}} + \hat{n} \quad (20)$$

$\dot{M}$ N m/s	model (17)+(18) °/s	of it (18) °/s	$\hat{n}$ °/s	cap (20) °/s	measured °/s	used
0.10	0.476	0.036	1.064	1.54	0.987	0.64
0.20	0.505	0.011	1.050	1.56	0.964	0.62
0.30	0.556	0.006	1.059	1.62	1.014	0.63
0.45	0.647	0.004	0.993	1.64	0.947	0.58
0.65	0.709	0.002	1.080	1.79	1.084	0.61
0.90	0.760	0.001	1.020	1.78	1.023	0.57
1.20	0.866	0.001	1.073	1.94	1.052	0.54

Inside on 140/140 runs; per-run used fraction p10 0.45, median 0.61, worst 0.86. On the exact nonlinear solution the realised δe_ω is 0.16–0.17°/s; the remaining factor ~ 3 of (17) over it is the 10° design box against the 4.8° the runs actually reach — the price of stating the bound before flying.

9 The two answers this gives

Why the residual is flat in $\dot{M}$. Nothing in (17) grows with the window — M_2 , M_1 are constants and the channel sups barely move — so the theory *predicts* a rate-flat residual, and the data agrees: the measured $\text{RMS}(r)$ stays at 0.95–1.08°/s across a twelvefold change in $\dot{M}$ (correlation with $\dot{M}$: $r = +0.09$, $p = 0.31$). The envelope (90), which grows like $\sinh x_{\text{end}}$, bounds e_ω — not the residual — and comparing the residual against it is what made the old cap look loose.

How far the shift moves the critical moment. The absorbed shift is (9), and the moment error it causes cancels $\dot{M}$ exactly:

$$|\Delta M_{\text{crit}}| = \dot{M} |\delta t_c| \approx \dot{M} \cdot \frac{e_\omega(\tau_{\text{end}})}{C_1 C_2 \sinh x_{\text{end}}} = \frac{W z_{CoM} e_\omega(\tau_{\text{end}})}{C_2 \sinh x_{\text{end}}} \leq \frac{W z_{CoM} E(\tau_{\text{end}})}{C_2 \sinh x_{\text{end}}} = \bar{\rho}, \quad (21)$$

using $\dot{M}/C_1 = W z_{CoM}$ and $E(\tau_{\text{end}}) = \bar{\rho} \sinh x_{\text{end}}/(J_P C_2)$. So the shift can displace the identified threshold by at most the window-averaged remainder — the same $\bar{\rho}$ the manuscript already budgets — and measured it uses 0.36–1.85 mN m of the ≈ 12 allowed, the sign lowering M_{crit} (earlier onset). (The chain’s middle step is the first-order $|\delta t_c|$; exactly, $|\delta t_c| = \text{artanh}(\beta/C_1)/C_2$, so the ceiling acquires a factor $\text{artanh}(u)/u$ with $u = \beta/C_1$: 1.0002 on the campaign (realised $u \leq 0.023$), up to 1.13 at the design-box corner of Sec. 9 where the a priori u of (9a) reaches 0.66.)

10 The bound over the design box

Because every constant in (17), (18) and (21) is available before the experiment, both results can be swept over the vehicle’s design box — mass 3.00–3.22 kg, $l_p = 0.110$ –0.140 m, CoM offset $p_{\text{off}} = \pm 0.020$ m (arm $= l_p - p_{\text{off}}$), $z_{CoM} = 0.20$ –0.30 m, $J_{CoM} = 0.050$ kg m² with J_P by the parallel-axis theorem, 600 combinations at each of the seven rates, each with its own tilt-limited window ($\sinh x_{\text{end}} - x_{\text{end}} = \varphi_{\text{max}} W z_{CoM} C_2 / \dot{M}$).

Two readings (Fig. 2). The residual bound stays within 0.26–1.00°/s everywhere in the box — below the measured disturbance floor of $\approx 1^\circ/\text{s}$ at almost every corner — so the fit-quality gate can be set from the noise figure alone for any vehicle in this class. And the identification error from the absorbed onset shift never exceeds 12.5 mN m anywhere in the box, at any rate — within 12% of the $\bar{\rho}$ budget at the worst corner and inside it everywhere else — with the same flatness in $\dot{M}$ that the campaign shows.

Reproduction

All numbers and figures in this document come from three scripts, run from the repository root:

```
python analysis/failing_runs.py          DataSet/exp    # writes the cache
python analysis/kernel_free_bound.py    docs/fig_kernel_free.png
python analysis/design_range_chart.py    docs/fig_design_range.png
```

`kernel_free_bound.py` evaluates (16)–(20) per run against the campaign cache and draws Fig. 1 (`M2()`, `M1()` implement (17a)–(17b); `model_term()` implements (16)–(18)). `design_range_chart.py` sweeps the design

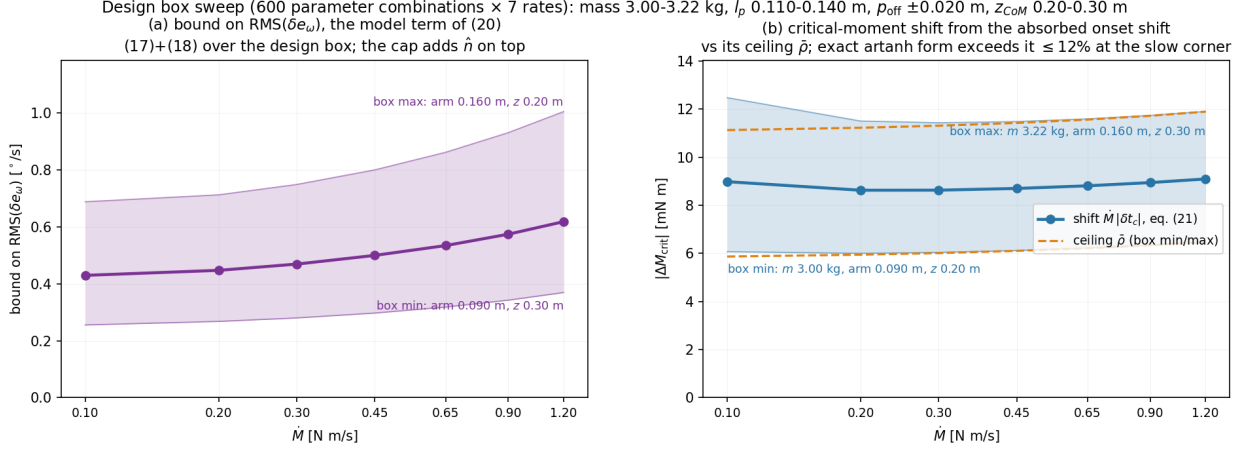


Figure 2: The design-box sweep. (a) the bound on $\text{RMS}(\delta e_\omega)$, eq. (17)+(18): $0.26\text{--}1.00^\circ/\text{s}$ over box and rates, median $0.43\text{--}0.62$. The band is set by the ratio arm/z_{CoM} — worst at arm 0.160 m with $z_{CoM} = 0.20$ m, best at arm 0.090 m with 0.30 m. (b) the critical-moment shift $M|\delta t_c|$ of (21): $6.0\text{--}12.5$ mN m , tracking its ceiling $\bar{\rho}$ and exceeding it by at most 12% at the slow-ramp corner — the $\text{artanh}(u)/u$ factor of the exact shift — while flat in $\dot{M}$ by the cancellation. Its worst corner has the *opposite* z direction — $m = 3.22$ kg, arm 0.160 m, $z_{CoM} = 0.30$ m — because $\bar{\rho}$ grows with the window that a higher CoM lengthens, while the residual bound worsens with arm/z_{CoM} .

box of Sec. 9 with the same closed forms and draws Fig. 2; its box limits are constants at the top of the file. Neither script solves an ODE or builds a kernel matrix — elementary functions and one scalar root-find (the window equation) per combination.

Remarks

(a) *Without the differential equation.* The same δe_ω can be written directly from the manuscript’s Duhamel form (89), giving $\delta e_\omega = \frac{1}{J_P} \int_0^{\tau_{\text{end}}} K(\tau, s) \rho(s) ds$ with a kernel that the addition identities collapse to $K = \sinh(C_2(\tau_{\text{end}} - \tau)) \cosh(C_2 s) / \sinh(C_2 \tau_{\text{end}})$ for $s \leq \tau$ (minus its mirror for $s > \tau$), and $\max |K| = 1$ exactly — the exponential cancels inside the kernel. It is an instructive identity, but as a bound it is weaker: K changes sign at $s = \tau$, so only $\int |K|$ can be used, and that costs a factor ≈ 2.4 over (17). The derivative bound in (16) is what forbids ρ from flipping sign with K , and that smoothness is worth exactly that factor.

(b) *What is a priori and what is not.* The model term of (20) uses only pre-experiment constants (list under (16)). The noise term (19) is measured: n_{hi} from the run itself, κ_q from quiet stretches, s_{med} one campaign median. A fully pre-experiment cap replaces $\hat{n}$ with a bench noise figure.

(c) *Exponent mismatch.* If the true exponent differs from the pinned C_2 by a relative ε_2 , the mismatch joins e_ω by definition and its unabsorbed part costs $0.018\text{--}0.022^\circ/\text{s}$ per 1% of ε_2 — $5\text{--}8\%$ of the model term, and the calibration residuals limit ε_2 to a few percent.

Appendix: the linear-algebra picture

Collect the N samples of a signal into a vector in $\mathbb{R}^N$. The model family (2) is then a subset of that space,

$$\mathcal{M} = \left\{ a(\cosh(c(\tau_i - d)) - 1) + b : a, c > 0, d, b \in \mathbb{R} \right\} \subset \mathbb{R}^N, \quad (\text{A.1})$$

a four-parameter curved manifold. Its tangent space at the nominal is spanned by the four derivative directions

$$T = \text{span} \left\{ \underbrace{\cosh(C_2 \tau) - 1}_{\partial/\partial C_1}, \underbrace{\sinh(C_2 \tau)}_{\partial/\partial \delta t_c}, \underbrace{\tau \sinh(C_2 \tau)}_{\partial/\partial C_2}, \underbrace{\mathbf{1}}_{\partial/\partial C} \right\}, \quad (\text{A.2})$$

so the family can move in at most four of the N dimensions around any of its points.

Why exact representation fails. “The true curve is a shifted nominal” asserts membership: $y_{\text{struct}} \in \mathcal{M}$. Two obstructions, matching the two in the Sec. 4 remark. Dimension: a four-dimensional manifold is a measure-

zero sliver of $\mathbb{R}^N$; a vector off it stays off it. Dynamics, sharper: every element of $\mathcal{M}$ samples a homogeneous solution of $\ddot{v} = c^2 v$, while y_{struct} samples the forced equation (12); by the argument under (13), membership would force $\dot{\rho} \equiv 0$. So y_{struct} has a component *transverse* to $\mathcal{M}$ that no parameter choice reaches.

What introducing δe_ω does. It changes the assertion from membership to a coordinate statement:

$$y_{\text{struct}} = m + \delta e_\omega, \quad m \in \mathcal{M}, \quad (\text{A.3})$$

exact for *every* choice of m — the freedom is in where to put the remainder, not whether there is one. Two canonical choices divide the labour:

choice of m	property of δe_ω	what it buys
orthogonal projection (LS)	$\delta e_\omega \perp T$, minimal norm	the smallest remainder, no pointwise structure
end-matched witness, (9)	$\delta e_\omega = 0$ at both window ends	the Dirichlet problem (13)–(14), hence (17)

The witness pays a factor 1.8–2.0 in norm over the orthogonal choice and buys, in exchange, the two boundary zeros that make the maximum principle — and with it the entire a priori bound — available. Provability is purchased with norm.

The projector sub-case, and the manuscript’s Appendix C. With C_2 pinned and the onset held, the family linearises to the subspace $\mathcal{V} = \text{span}\{u, \mathbf{1}\}$, $u_i = \cosh(C_2 \tau_i) - 1$, and the orthogonal choice becomes an explicit matrix: with $A = [u \ \mathbf{1}]$,

$$P = A(A^\top A)^{-1} A^\top, \quad P_{ij} = \frac{1}{N} + \frac{\tilde{u}_i \tilde{u}_j}{\sum_k \tilde{u}_k^2}, \quad r_\perp = (I - P)(e_\omega + n), \quad (\text{A.4})$$

($\tilde{u} = u - \bar{u}$) — the manuscript’s Lemma 2, eq. (C3). The δe_ω of this document is the same object one level up: the remainder transverse to the full nonlinear family rather than to its pinned linearisation.

How much of e_ω lies along the family. Measured on the exact tip-over, the fraction of e_ω (in RMS) absorbed by successively larger reachable sets: 61–67% by $\text{span}\{u, \mathbf{1}\}$, 70–71% adding the onset direction $\sinh$, 92.5% by the full four-parameter family — the last flat across all rates. The transverse component — the 7.5% no parameters reach, 0.012–0.018°/s — is the irreducible core of δe_ω , and (17) is its a priori ceiling.